

Inventory Risk Management & Decision Support System

AI-assisted platform for preventing stockouts, reducing overstock, and improving inventory decisions

THE PROBLEM

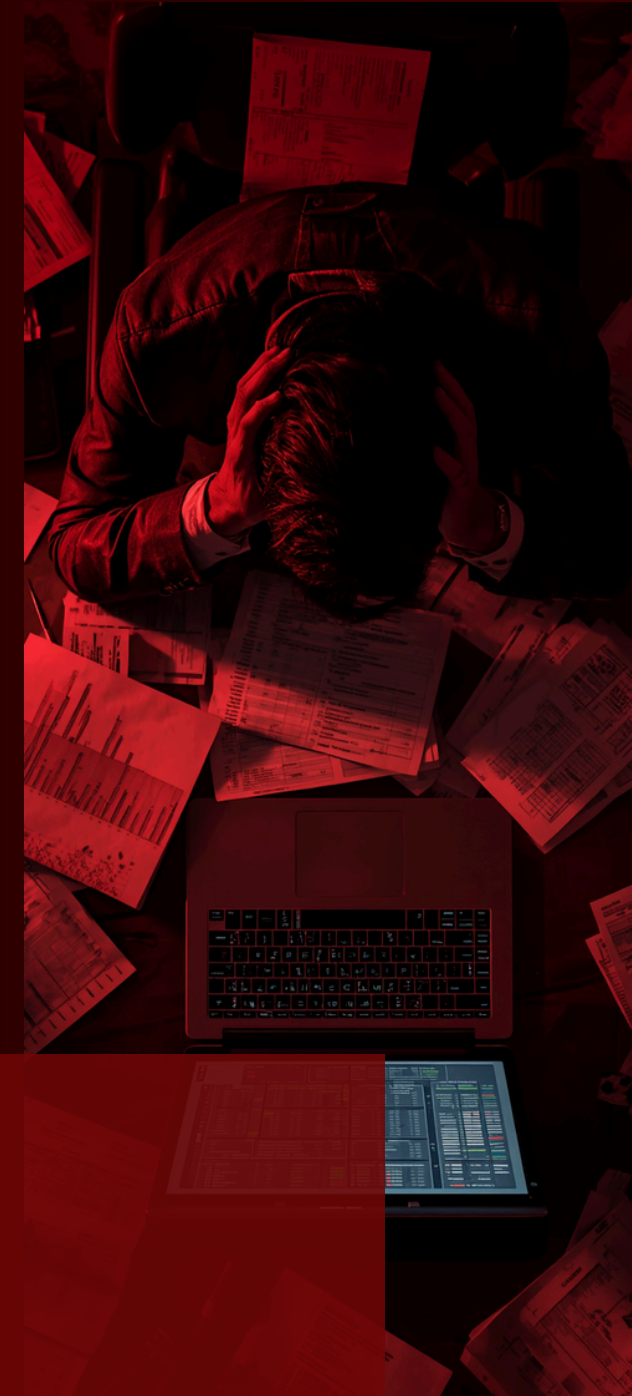
Inventory teams react too late to stockout and overstock risk

Manual monitoring across SKUs, suppliers, and sales documents creates delays in detecting stock issues. Teams struggle to keep up with constantly changing demand patterns and supplier lead times.

Manual monitoring is slow, error-prone, and costly

Poor inventory decisions lock up working capital in slow-moving goods and hurt service levels through stockouts. Small and mid-sized teams often lack the intelligent decision-support tools needed to stay ahead.

Many businesses still rely on spreadsheets or fragmented systems, causing missed sales and tying up cash in excess inventory.





Teams lack a unified view of inventory risk, supplier risk, and transaction history.



Restocking decisions are manual and based on incomplete data signals.



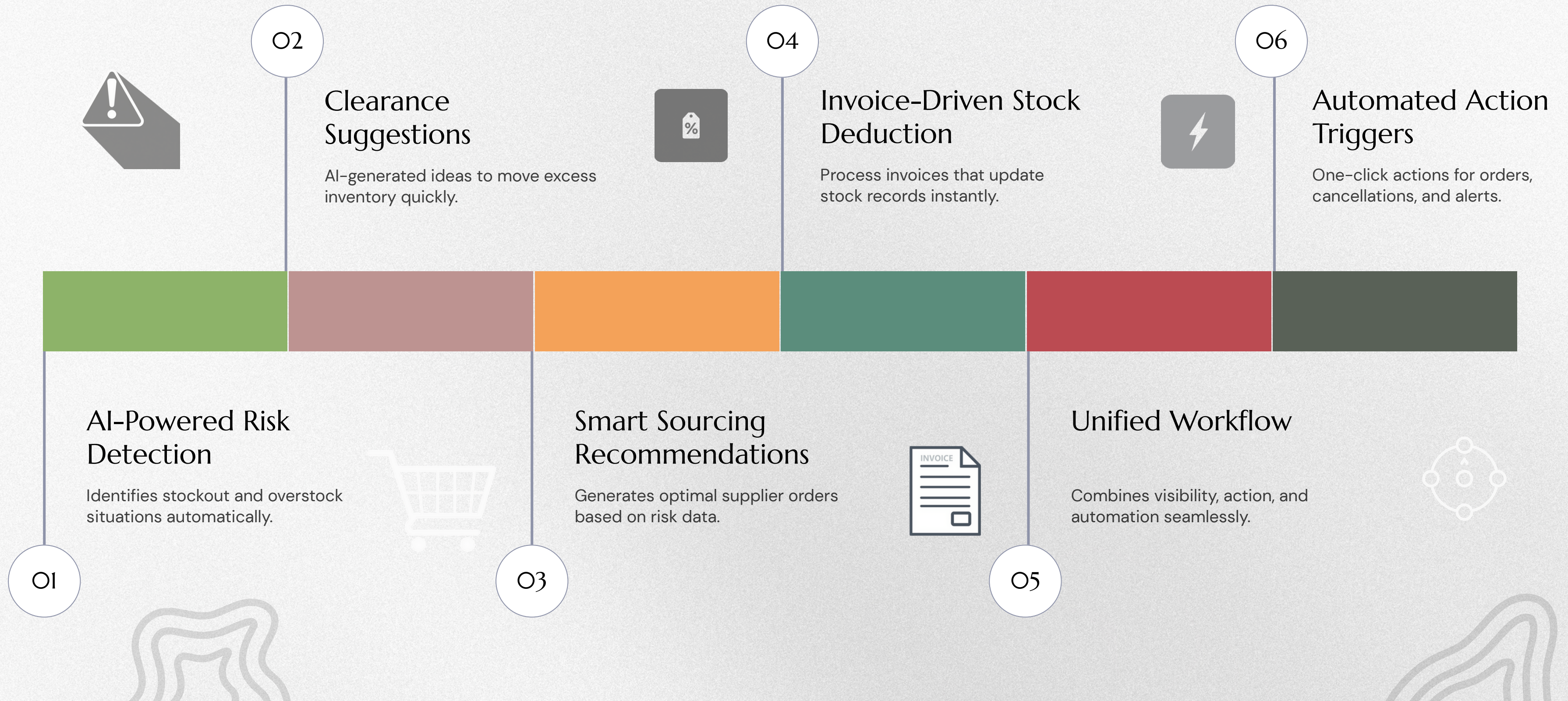
Excess stock remains unsold without actionable recovery strategies.



Current Pain Points

OUR SOLUTION

Full-stack inventory risk dashboard with AI-assisted recommendations



Product Demo Flow



Dashboard Login

User logs in and enters the inventory dashboard to access real-time stock data

Urgent SKU Alerts

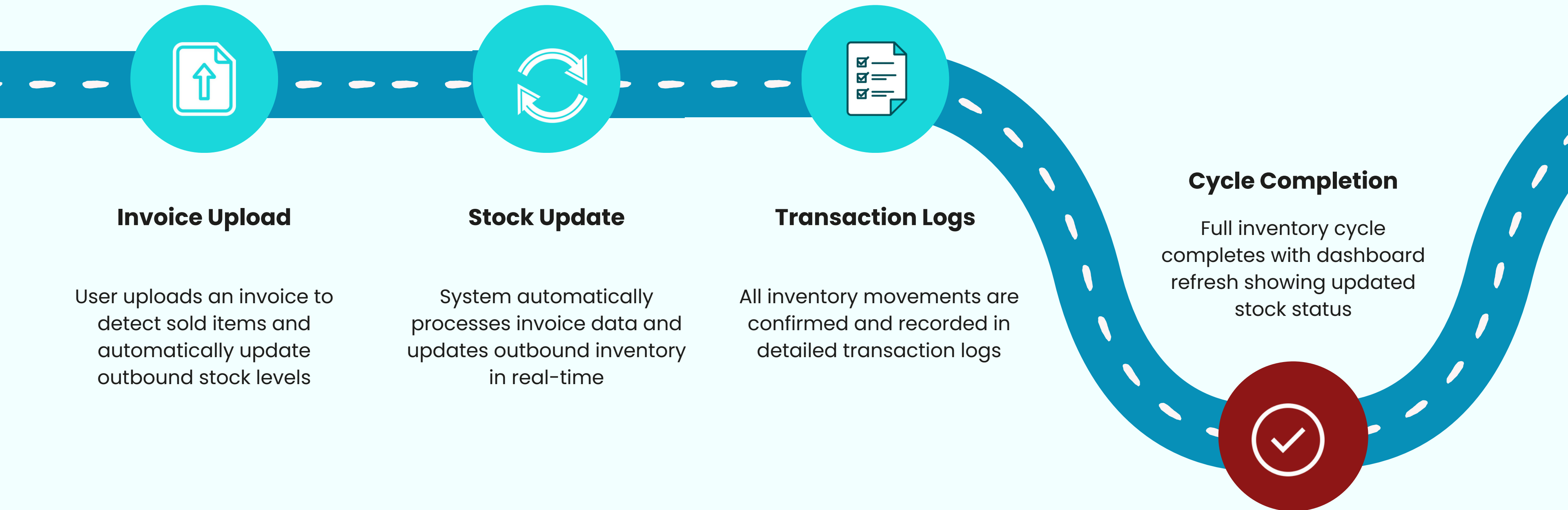
System highlights urgent SKUs using a 15-day stockout forecast for proactive planning

AI Recommendations

User reviews sourcing suggestions or runs AI clearance recommendations for optimization



Product Demo Flow



SYSTEM ARCHITECTURE

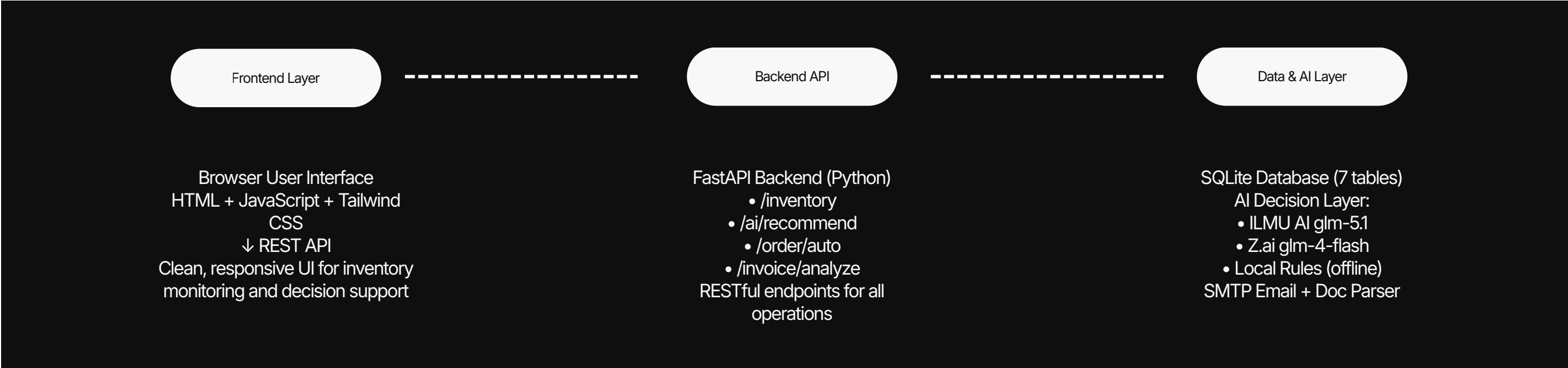
Frontend Layer
Static HTML pages with JavaScript for dynamic interactions. Tailwind CSS provides responsive styling. Dashboard components handle real-time inventory displays and user controls.



Backend Services
FastAPI service exposing RESTful endpoints for inventory management, reporting, supplier operations, authentication, AI integration, and invoice processing APIs.



Database Layer
SQLite database with structured tables for inventory items, transactions, supplier records, user accounts, and workflow management. Lightweight and portable.



Key Features

Our inventory management system delivers five powerful capabilities designed to streamline operations, reduce risk, and maximize efficiency across your entire supply chain.



Inventory Risk Dashboard

Get instant visibility into stock levels, days of inventory remaining, and urgent items requiring immediate action. Make faster, smarter decisions with real-time data at your fingertips.



AI Clearance Strategy

Leverage AI-powered analysis to generate effective liquidation strategies for slow-moving inventory. Turn excess stock into revenue with data-driven clearance recommendations.



Transaction Reporting

Maintains a complete, visible audit trail of all stock movements. Every transaction is logged and easily accessible, ensuring transparency and accountability across your entire inventory operation.



Smart Sourcing Workflow

Receive intelligent recommendations for low-stock replenishment with support for automated ordering. Streamline procurement and never miss a restocking opportunity.



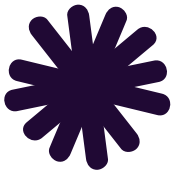
Invoice Scanner

Automatically process PDF, Word, and Excel invoices to extract and infer sold items. Reduce manual data entry and improve accuracy across your transaction records.



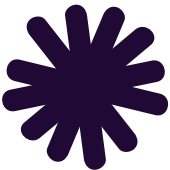
Why This Is Different

Our solution combines detection, recommendation, and operational action in one unified interface. Unlike generic AI tools, it's designed around practical inventory pain points, linking inventory risk directly to supplier decisions rather than just reporting.



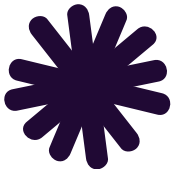
Unified Interface

Detection, recommendation, and operational action combined in a single streamlined interface for faster decision-making.



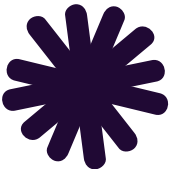
Risk-to-Action Linking

Connects inventory risk directly to supplier decisions, going beyond simple reporting to enable proactive management.



Multi-Format Ingestion

Supports diverse document formats for real operational workflows, adapting to how your team actually works.



Built-In Resilience

Includes fallback logic ensuring demo stability even when AI services are unstable, designed for practical use.



WHY THIS IS DIFFERENT



See

Detection identifies inventory risks automatically, scanning data patterns and anomalies across your entire supply chain in real-time.

Decide

AI-powered recommendations analyze risk factors and suggest optimal actions, prioritizing decisions based on business impact and urgency.

Act

Execute operational decisions directly from the interface. No switching between systems—take action immediately on recommendations.

Integrate

Link inventory risk directly to supplier decisions. Not just reporting—actual workflow integration with your supplier network.

Ingest

Multi-format document processing handles invoices, POs, and supplier docs. Real operational workflows, not generic chatbot interactions.

Impact and Roadmap

Short-term Impact

Reduce manual effort and improve response speed to inventory risk.



Business Value

Protect cash flow, reduce stock waste, and avoid missed sales.



1 Month Goals

Add charts, better auth, invoice parsing, role-based access.



1 Week Goals

Stabilize deployment, fix config issues, and harden demo flow.



3 Month Goals

Forecasting models, supplier scoring, multi-user cloud deployment.





77Spectre

CONCLUSION

"From Inventory Visibility to Actionable Decisions"

Inventory decisions should be proactive, not reactive. Our system turns inventory data into clear operational action. Built as a working prototype with AI-assisted workflows and real demo paths.

GitHub: github.com/Jiajun3103/Inventory-Risk-Management-and-Decision-Support-System-77Spectre.git

Demo Video: https://drive.google.com/file/d/1rvrKyKJ5ypVVMvl0LtdVZSaFa7gU7ofS/view?usp=drive_link

